

Responsible for Q1, Q3

a) Report the validation accuracy of User-based KNN.

I evaluated the validation accuracy of the user-based KNN of the following K values: 1, 6, 11, 16, 21, 26. The corresponding accuracies are 0.6250, 0.6785, 0.6888, 0.6754, 0.6685, 0.6509.

b) The K value that gives the highest performance on validation data (< 0.6888) is 11. The final test accuracy on $k=11$ is 0.6833.

c) I evaluated the validation accuracy of the item-based KNN using the following value of K: 1, 6, 11, 16, 21, 26. The corresponding validation accuracies are 0.6171, 0.6569, 0.6805, 0.6901, 0.6907, and 0.6902 respectively.

For user-based KNN, the best validation accuracy is 0.6907 with $k=21$. Which means using the 21 most similar questions produced the best performance on the validation dataset. The same K value gives the highest test accuracy on item-based KNN, which is 0.6830.

(d)

The user-based KNN achieved a test accuracy of 0.6833 with $K=11$. The item-based KNN achieved a test accuracy of 0.6833 with $K=21$. Although the validation accuracy of item based KNN is slightly higher than the validation accuracy of user-based KNN, then user-based KNN performs slightly better on the test set. Therefore, user-based KNN performs better.

(e)

First limitation: KNN is sensitive to sparse data. Since many students' response are missing, the similarity calculations may be less reliable, which lowers the accuracy. The second limitation: KNN is computationally expensive. Which means it is less efficient on large datasets.

Q3 (Choose Neural Network)

(a)

Difference 1: ALS is a matrix factorization method which uses low dimensional latent vectors to model students and questions.

Neural network uses multiple layers of non-linear transformation.

Difference 2: ALS alternates least-square updates in order solve optimization problem. Neural network is trained using gradient based optimization.

Difference 3: ALS is a linear factorization model. Capturing patterns in the data can be hard. Neural network is able to model non-linear

relationships between students and questions

Question 3 (ii)

(d) The autoencoder was trained using different latent dimensions, learning rates, and number of epochs. The best model used $k=50$, a learning rate of 0.01, and 50 epochs, achieving a validation accuracy of 68.40% and a test accuracy of 68.56%. The training and validation objectives both decreased during training, which means the learning process is efficient. The validation objective leveled off after about 35 epochs, suggesting that additional training provided only small improvements.

(e)

We added L2 regularization to the training objective and tested $\lambda \in \{0.001, 0.01, 0.1, 1\}$. The best result was obtained with $\lambda = 0.001$, giving a validation accuracy of 68.97%, which is slightly higher than the 68.40% without regularization. Larger values of λ reduced the validation accuracy, indicating overfitting. Therefore, a small amount of L2 regularization improved the model, with stronger regularization decreasing performance.

2. item Response Theory

(a)

Let O denote the set of observed student-question pairs. For student i and question j , the one-parameter IRT model defines the probability of a correct response as

$$p_{ij} = P(C_{ij}=1 \mid \theta_i, \beta_j) = \sigma(\theta_i, \beta_j) = \sigma(\theta_i - \beta_j) = \frac{e^{\eta(\theta_i - \beta_j)}}{1 + e^{\eta(\theta_i - \beta_j)}}$$

where θ_i represents the ability of student i , and β_j represents the difficulty of question j .

Since each response c_{ij} is binary, the likelihood of one observ

$$P(C_{ij} \mid \theta_i, \beta_j) = p_{ij}^{c_{ij}} (1 - p_{ij})^{1 - c_{ij}}$$

$$P(C \mid \theta, \beta) = \prod_{(i,j) \in O} p_{ij}^{c_{ij}} (1 - p_{ij})^{1 - c_{ij}}$$

$$\log_e P(C \mid \theta, \beta) = \sum_{(i,j) \in O} [c_{ij} \log p_{ij} + (1 - c_{ij}) \log (1 - p_{ij})]$$

gradient with respect to θ_i is

$$\frac{\partial \log P(C \mid \theta, \beta)}{\partial \theta_i} = \sum_{j=(i,j) \in O} [c_{ij} - p_{ij}]$$

gradient with respect to θ_j is

$$\frac{\partial \log P(c|\theta, \beta)}{\partial \theta_j} = \sum_{i: (i,j) \in O} (p_{ij} - c_{ij})$$

(b)

We initialized all student ability parameters and question difficulty parameters to zero. Therefore, the initial predicted probability for every response was 0.5. We used gradient ascent to maximize the log-likelihood

$$\theta_i \leftarrow \theta_i + \eta \sum_{j: (i,j) \in O} (c_{ij} - p_{ij})$$

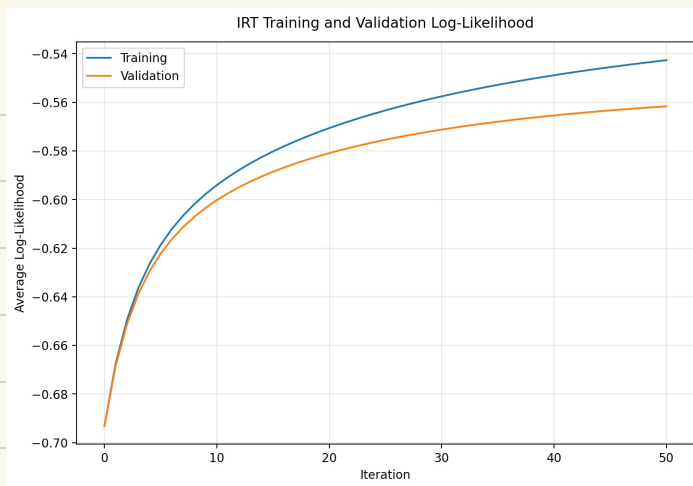
$$\beta_j \leftarrow \beta_j + \eta \sum_{i: (i,j) \in O} (p_{ij} - c_{ij})$$

where η is learning rate

we tested learning rates in $\{0, 0.01, 0.0025, 0.005, 0.01, 0.02, 0.05\}$

and compared different numbers of iterations using the validation set. We selected a learning rate of 0.0025 and 50 iterations.

Figure 1 shows the average training and validation log-likelihood per observed response. We used the average because the training and validation sets contain different numbers of responses. Both curves increased during training, showing that the model learned the student ability and question difficulty parameters. The training log-likelihood was slightly higher than the validation log-likelihood, which is expected because the model was fitted using the training data. The validation curve did not decrease, so there was no clear evidence of overfitting during the selected training period



(c) using learning ratio of 0.0025 and 50 training iterations, the final results were

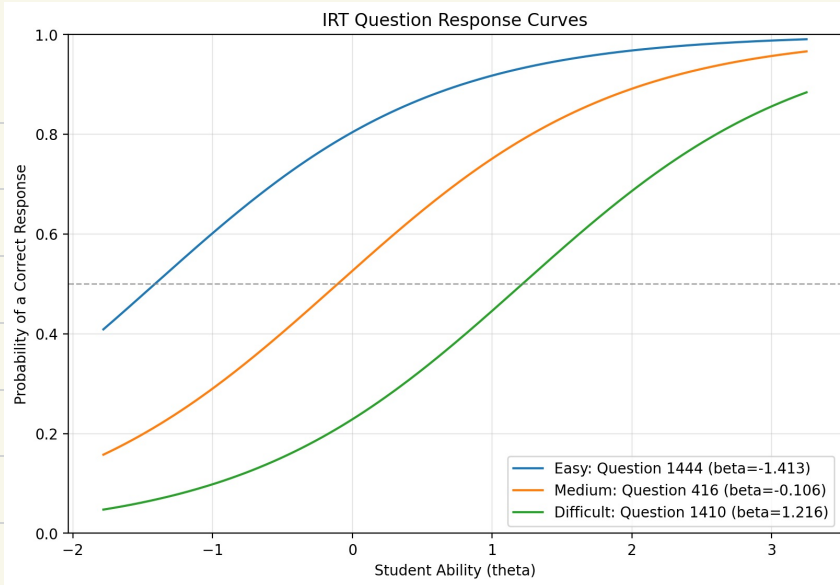
	accuracy
Validation	0.7084
Test	0.7005

The validation accuracy was 70.84%, while the test accuracy was 70.05%. The small difference suggests that the IRT model generalized reasonably well to unseen responses

(d)

We selected three questions using their learned difficulty parameters. We chose the question with the lowest β , a question with a median β , and the question with the highest β

question	Difficulty level	β_j
1444	easy	-1.4133
416	medium	-0.1059
1410	difficult	1.2158



All three curves are sigmoid and increase with student ability. The easy question is shifted to the left, whereas the difficult question is shifted to the right. Since $p(c_{ij} = 1) = 0.5$ when $\theta = \beta_j$, each curve's 0.5 crossing identifies the learned question difficulty. Their slopes are the same because the one-parameter IRT model does not learn a separate discrimination parameter

4 Bagging Ensemble

We generated three bootstrap samples from the original 56,688 training responses, using replacement and random seeds 1, 2, and 3. One IRT model was trained on each sample with learning rate 0.0025 for 50 iterations. Repeated and omitted responses in the samples caused the three models to learn different θ and β parameters.

For student-question pair (i, j) , base model b produced

$$p_{ij}^{(b)} = \sigma(\theta_i^{(b)} - \beta_j^{(b)}).$$

$$\bar{p}_{ij} = \frac{1}{3} \sum_{b=1}^3 p_{ij}^{(b)}$$

the final prediction was

$$\hat{c}_{ij} = \begin{cases} 1, & \bar{p}_{ij} \geq 0.5 \\ 0, & \bar{p}_{ij} < 0.5 \end{cases}$$

Model	validation accuracy	Test accuracy
bootstrap 1	0.6973	0.6870
2	0.7007	0.7025
3	0.6976	0.6901
ensemble	0.7045	0.6991
original full-data	0.7084	0.7005

The ensemble's validation accuracy of 0.7045 exceeded all three bootstrap models, suggesting that averaging reduced some sampling variance. Its test accuracy of 0.6991, however, was below Bootstrap IRT 2 (0.7025) and the original full-data IRT model (0.7005), so the gain was not consistent. Bootstrap samples repeat some responses and omit others, giving each model less unique information than full-data training. Moreover, the three IRT models make correlated errors. Bagging can reduce variance, but it cannot remove the bias or underfitting of the one-parameter IRT model. Thus, the ensemble was more stable than its bootstrap components but did not improve the final test result

Part B

(1)

Baseline limitation. The one-parameter logistic (1PL) model from Part A predicts student (i)'s response to question

(j) as

$$p_{ij} = p(c_{ij} = 1 \mid \theta_i, \beta_j) = \sigma(\theta_i - \beta_j).$$

Here, θ_i is student ability and β_j is question difficulty. This model forces every question response curve to have the same slope. That assumption is restrictive: some questions strongly separate students just above and below the required ability, while other questions are noisy and provide little information about ability.

Proposed extension. We replace the 1PL model with the standard two-parameter logistic (2PL) model [1]:

$$p_{ij} = \sigma(z_{ij}), \quad z_{ij} = a_j (\theta_i - \beta_j).$$

The new discrimination parameter $a_j > 0$ controls the slope of question j's response curve. A large a_j produces a

steep transition near $\theta_i = \beta_j$; a small value produces a flatter, less informative curve. To enforce positivity, we optimize an unconstrained parameter y_j and set

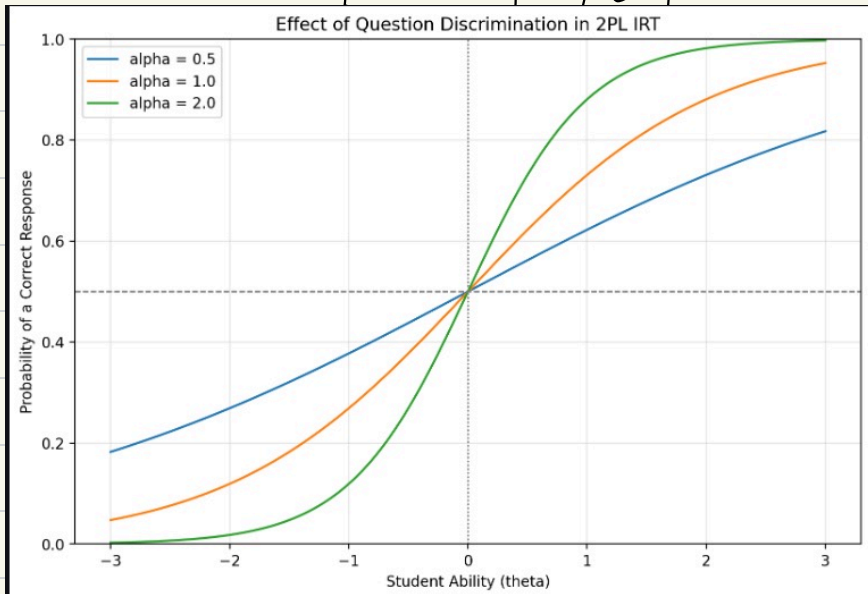
$$a_j = \exp(y_j).$$

We initialize $y_j = 0$, hence $a_j = 1$, so training begins exactly at the original 1PL assumption.

For observed responses O , we maximize a regularized log-likelihood

$$J(\theta, \beta, \gamma) = \sum_{i,j: (i,j) \in D} [c_{ij} z_{ij} - \log(1 + \exp z_{ij})] - \frac{\lambda}{2} (\|\theta\|_2^2 + \|\beta\|_2^2 + \|\gamma\|_2^2).$$

L^2 penalty is intended to reduce overfitting caused by the additional question-specific parameter



2. let $r_{ij} = c_{ij} - p_{ij}$. Applying the chain rule to the objective gives the data-dependent gradient

$$\frac{\partial J}{\partial \theta_i} = \sum_{j: (i,j) \in D} d_j r_{ij} - \lambda \theta_i$$

$$\frac{\partial J}{\partial \beta_j} = \sum_{i: (i,j) \in D} -d_i r_{ij} - \lambda \beta_j$$

$$\frac{\partial J}{\partial r_{ij}} = \sum_{i: (i,j) \in D} d_i (\theta_i - \beta_j) r_{ij} - \lambda r_{ij}$$

All parameters are updated simultaneously by gradient ascent. We clipped γ_j to $[-3, 3]$ only as a numerical safeguard, which restricts α_j to approximately $[0.05, 20.09]$. A finite-difference gradient check on a synthetic dataset gave a maximum absolute error of 4.4×10^{-10} .

Training algorithm. (1) Initialize $\theta = 0$, $\beta = 0$, $\gamma = 0$. (2) Compute $\alpha = \exp(\gamma)$, probabilities, and residuals. (3) Accumulate the three gradients over observed responses. (4) apply L2 gradients and update the parameters. (5) retain the checkpoint with the highest validation accuracy.

Controlled experiment. The data contain 56,688 training, 7,086 validation, and 3,543 public-test responses. We trained every configuration for at most 100 full-batch updates. We evaluated the Cartesian grid $\eta \in \{0.001, 0.0025, 0.005, 0.01\}$, $\lambda \in \{0, 1, 3, 10\}$. Initialization, data, update equations, and the maximum number of iterations were held fixed. For each configuration, the checkpoint was selected using validation accuracy. The test set was not used during model selection; it was evaluated only after the best validation configuration had been fixed.

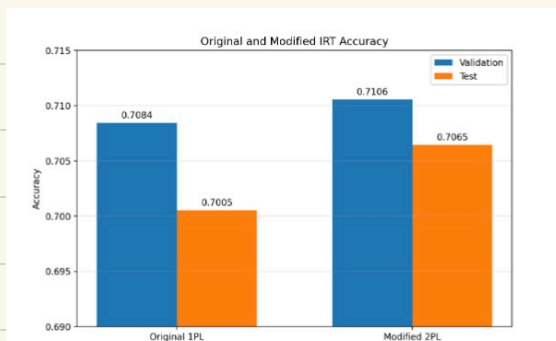
This experiment tests two hypotheses. First, learning α_j should help if the equal-slope assumption underfits the response patterns. Second, moderate regularization should improve generalization, while very strong regularization should push the parameters toward zero and cause overfitting

2PL validation accuracy

learning rate	0.001	0.7087	0.7090	0.7090	0.7076
	0.0025	0.7096	0.7091	0.7094	0.7094
	0.005	0.7099	0.7099	0.7093	0.7099
	0.01	0.7096	0.7106	0.7097	0.7099

3. The best validation setting was $\eta = 0.01$, $\lambda = 1$, with checkpoint 69. The learned discriminations had minimum 0.3235, median 1.3690, and maximum 6.0441. Since these values are not all near one, the modified model used the extra flexibility rather than collapsing back to 1PL

model	η	λ	validation	Test
original	0.0025	-	0.7084	0.7005
2PL, no L2	0.005	0	0.7099	-
selected 2PL	0.0100	1	0.7106	0.7065



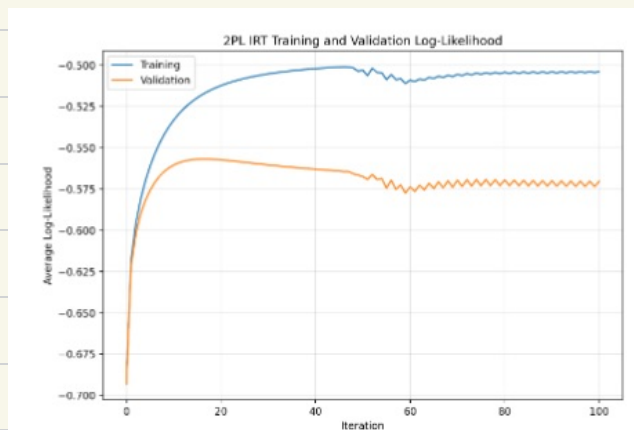
The modified model increased validation accuracy by 0.0021 (0.21 percentage points) and test accuracy by 0.0059

(0.59 percentage points). The improvement is small but consistent across the two held-out sets. This supports the hypothesis that fixed discrimination was a source of underfitting.

The grid also gives limited evidence for the regularization hypothesis. At $\eta = 0.01$, validation accuracy rose from 0.7096 with $\lambda = 0$ to 0.7106 with $\lambda = 1$, then fell to 0.7097 with $\lambda = 3$. Across the grid, results were close, indicating that regularization was not the only cause of the improvement. Learning rate and early checkpoint selection also mattered.

The selected model's training accuracy was 0.7408, substantially above its validation accuracy of 0.7106. Its average

training NLL was 0.5069, compared with validation NLL 0.5739. These gaps show that the additional parameters still introduce overfitting even though classification accuracy improved



Sparse questions and overfitting. The model adds one parameter per question. For questions with few observed responses, a_j can be estimated from very little evidence and may reflect noise. The wide learned range (0.3235 to 6.0441) suggests that some estimates are unstable. A hierarchical prior or stronger question-dependent shrinkage could

pull poorly supported discriminations toward one.

Identifiability and optimization. Ability, difficulty, and discrimination can trade off in scale, making 2PL less identifiable and harder to optimize than 1PL. The selected learning rate produced visible oscillation in held-out likelihood. Better constraints, mini-batch optimization, Adam, or a decaying learning rate could improve stability.

The current clipping interval is also a practical safeguard rather than a theoretically selected constraint.

One-dimensional ability. Both 1PL and 2PL compress every student into one scalar θ_i . A student may be strong in algebra but weak in geometry, so a single ability cannot model subject-specific knowledge. A multidimensional IRT model could use question_meta.csv subject labels and learn one ability per subject.

Missing response mechanisms. The model does not represent guessing, careless mistakes, or ambiguous questions.

A 3PL model could add a lower-asymptote guessing parameter, although it would add further estimation difficulty.

Student and question metadata could also provide features for users or items with few observations.

Metric mismatch. The best checkpoint was selected by 0.5-threshold accuracy, while the validation log-likelihood had already begun to worsen. Accuracy ignores probability calibration. Future evaluation should report NLL or Brier score together with accuracy and use a predeclared selection metric.

Conclusion. Allowing item-specific discrimination produced a modest improvement over the Part A baseline, supporting the claim that equal slopes caused some underfitting. However, the small gain, generalization gap, and optimization sensitivity show that added flexibility must be controlled carefully

Team Contributions

bo was primarily responsible for Part A Question 1 (K-Nearest Neighbours) and Question 3 (Matrix Factorization).

Jincheng Yang was primarily responsible for Part A Question 2 (Item Response Theory), Question 4 (Bagging Ensemble), and Part B, including the 2PL IRT extension, experiments, figures, and analysis.

Both team members reviewed each other's code, experimental results, and written sections. [Teammate Name] also reviewed the Part B implementation and analysis, while Jim Yang reviewed the work for Questions 1 and 3. Both members participated in the final editing, verification, and integration of the report.

Course Evaluation Confirmation

We confirm that all members of the team have completed and submitted the CSC311 course evaluation.

